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Research Paper Title:

Helping Hands: A Machine Learning Approach to Prioritizing NGO Assistance

Track

This paper falls under the track of Humanitarian Technology and ICT for Social Good, as it focuses on leveraging digital platforms to enhance transparency, efficiency, and inclusivity in community aid distribution

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Helping Hands: A Machine Learning Approach to Prioritizing NGO Assistance

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Abstract-

Despite growing willingness to contribute towards social causes, a persistent gap exists between donors and organizations in need due to limited visibility, communication barriers and lack of trust. Helping hands is a web-based humanitarian aid platform designed to reduce this gap by enabling direct, transparent and efficient interaction between donors, volunteers and Non-Governmental Organizations (NGOs). The platform allows NGOs and community helpers to post real-time requests for resources such as food, clothing, medical supplies and educational support, while donors can easily discover opportunities aligned with their intent to help.

A key innovation of this system is the integration of machine learning techniques to improve decision-making and resource allocation. Location-aware recommendation models suggest nearby NGOs to donors, ensuring quicker response times and improved engagement. Additionally, request prioritization based on urgency and resource type enhances visibility for the critical needs. By combining real-time data handling, personalization and intelligent recommendations, Helping Hands aims to create a reliable and inclusive ecosystem where assistance is delivered efficiently and ethically to those who need it the most.

Keywords: humanitarian aid platform, NGOs, machine learning, transparency, resource prioritization.

1.INTRODUCTION

Humanitarian organizations play a vital role in supporting marginalized and vulnerable populations by providing essential resources and services. Despite their importance NGOs-especially small and community-driven initiatives, often face operational challenges that limit their effectiveness. These challenges include inadequate outreach, limited digital presence, inefficient communication channels and difficulty in attracting consistent donor support. On the other hand, individuals and organizations willing to contribute frequently lack reliable information about whether their assistance is most needed, and how will it be utilized.
This disconnect between aid providers and aid seekers results in delayed responses, inefficient use of resources and in some cases, unmet urgent needs.

The problem becomes more severe during emergency, such as natural disasters, health crisis or economic downturns, and where timely assistance is critical. Traditional aid distribution mechanisms are often slow, fragmented and depend on mutual coordination.

The Helping Hands platform is designed to address such issues by acting as a centralized digital bridge between donors, volunteers and NGOs by enabling real-time request submissions, urgency-based prioritization and location-aware recommendations. The platform seeks to improve response times and enhance transparency. The system emphasizes inclusivity by supporting small and grassroot organizations alongside larger NGOs, thereby strengthening the overall humanitarian ecosystem

II. LITERATURE SURVEY

Several digital platforms have been developed to improved aid distribution. However, each presents notable limitations. Platform such as GoFunfMe me and Ketto are effective in mobilizing large-scale donations, but often prioritize high-visibility campaigns, making it difficult for small NGOs and local initiatives to gain exposure [1]. Existing NGO management systems focus on financial transparency and are typically designed for large organizations, leaving smaller groups and individual helpers undeserved [2].
Volunteer platforms such as VolunteerMatch efficiently connects volunteers to opportunities lack real-time urgency indicators and direct aid request mechanisms limiting their effectiveness during the emergency[3]. Trust and transparency remain as significant challenge in digital aid ecosystems, as donors are often uncertain whether their contributions reach towards the intended beneficiaries[4]. While blockchain-based models have been proposed to enhance transparency, their complexity restricts adoption among grassroots organizations[5].
Real-time dashboards have been proven valuable in humanitarian logistics and disaster response; however, such tools are mostly accessible to government bodies and large NGOs leaving smaller organizations without adequate technological support[6]. *Helping Hands* seeks to address these gaps such through an inclusive, lightweight and intelligent platform.

III. MACHINE LEARNING METHODOLOGY

Flow:

Data Preparation
↓
Similarity Scoring
↓
Model Training
↓
System Output

Step 1: Data Preparation
Convert NGO locations to latitude-longitude coordinates (geocoding).
Standardize urgency levels numerically:
High = 3, Medium = 2, Low = 1.
Clean and normalize other features, including NGO type, help type, and user location.

Step 2: Similarity & Scoring
Calculate the distance between the user's location and each NGO using the Euclidean method.
Score each NGO by combining distance and urgency score:

$$\text{NGO Score} = w_1 \cdot \left(\frac{1}{\text{distance}_{\text{user, ngo}} + 1}\right) + w_2 \cdot \left(\frac{\text{urgency}}{3}\right)$$

w1 and w2 are weights that balance distance and urgency.

The formula gives higher scores to NGOs that are both closer and more urgent.
Adjusting w1 and w2 (for example, w1 = 0.5, w2 = 0.5) changes the focus between proximity and urgency.

Step 3: Model Training
Use k-Nearest Neighbors (kNN) or Decision Tree as baseline models for recommendations.
If the dataset is large, consider using neural networks or ensemble models for non-linear weighting.

Step 4: System Output
Recommend the top N NGOs with the highest scores.
Update recommendations as the user's location or urgency profile changes.
Example to check the working

$$\text{NGO Score} = 0.5 \cdot \left(\frac{1}{\text{distance}_{\text{user, ngo}} + 1}\right) + 0.5 \cdot \left(\frac{\text{urgency level}}{3}\right)$$

distance user, ngo is calculated using latitude and longitude via the Haversine or Euclidean formula.
urgency level is 1, 2, or 3 for Low, Medium, and High, respectively.
The highest-scoring NGOs are recommended to the user.

Workflow of the model is
- Location Geocoding
- Urgency Encoding
- Score Calculation
- Recommendation Output

Algorithm used in this model:

The recommended algorithm for suggesting the nearest and most needful NGO is the weighted k-Nearest Neighbors (kNN) algorithm with a custom scoring function that combines both spatial distance and urgency.

Weighted k-Nearest Neighbors Explained:
Dataset Preparation
- Prepare a dataset where each NGO entry includes latitude, longitude, urgency level (converted to numeric weight), and needed help type.

Distance Calculation
- For each NGO, calculate its distance to the user's location.
  - The Euclidean distance formula is:

$$d_i = \sqrt{(x_{\text{ngo}_i} - x_{\text{user}})^2 + (y_{\text{ngo}_i} - y_{\text{user}})^2}$$

  - where (x,y)(x,y) are latitude and longitude coordinates of NGO ii and the user.

Urgency Weighting
- Assign a numerical value to urgency: Low = 1, Medium = 2, High = 3. You can further scale urgency importance.

Score Calculation
- Instead of simple neighbor majority, use a weighted kernel **function** for each NGO:

$$\text{score}_i = w_1 \cdot \frac{1}{d_i + \epsilon} + w_2 \cdot \text{urgency}_i$$

Where:

- d_i: Distance from user to NGO i
- ϵ: Small constant to avoid division by zero (e.g., 0.001)
- w_1, w_2: Weights controlling importance of distance/urgency (set based on use-case, e.g., $w_1 = 0.5, w_2 = 0.5$)
- urgency_i: Numeric urgency value.

Algorithm Steps:
a. For the user's location, calculate distances to all NGOs.
b. Choose the kk NGOs with the highest weighted scores.
c. Recommend those NGOs to the user.

Full Algorithm Outline
1. Input: User location (x,y)(x,y), list of NGOs with location and urgency.
2. For each NGO:
   - Compute Euclidean distance didi
   - Compute urgency score (numeric value)
   - Calculate:

$$\text{score}_i = w_1 \cdot \frac{1}{d_i + \epsilon} + w_2 \cdot \text{urgency}_i$$

1. Sort NGOs by score.
2. Return top kk NGOs.

Implementation Notes:

- Any decreasing kernel function can be used (common: inverse distance).
- Can blend with classification/regression for predicting other attributes.
- Works well for location-based services where urgency matters

IV. SYSTEM REQUIREMENTS

Technology Stack

- Frontend: HTML, CSS (Bootstrap)
- Backend: Flask (Python)
- Database: Google Sheets for live data
- Integration: Google Forms for submission, Pandas for data fetching and manipulation

V.WORKFLOW

- Requests are submitted through Google Forms.
- Data is centralized in Google Sheets.
- The backend fetches live data for the dashboard.
- Supporters browse verified needs and can respond instantly.

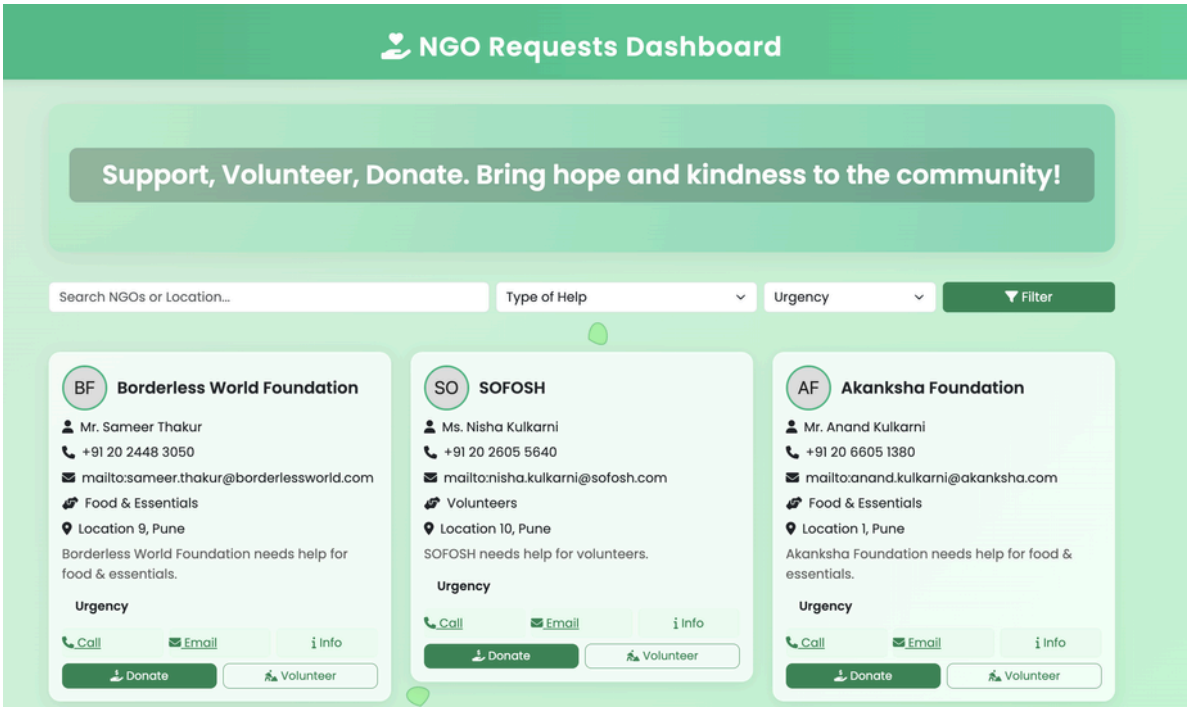
Google Form link:

The link will help in collecting the information from NGOs

https://docs.google.com/forms/d/e/1FAIpQLScGOFQgDoZItWoKl2jZfXNbmldNQENiVxHGZknMFrqUp_qWkg/viewform?usp=header

-Website:

This is an NGO Requests Dashboard website that helps donors and volunteers connect with NGOs based on their needs. The layout is clean, simple, and easy to use with filters for location, type of help, and urgency.



Functions:

1. Search Bar – Allows users to search NGOs by name or location.
2. Type of Help Filter – Lets users choose the type of support needed (e.g., food, essentials, volunteers).
3. Urgency Filter – Sorts NGOs based on urgency levels (High, Medium, Low).
4. NGO Cards – Display NGO details such as name, contact info, email, location, type of help needed, and urgency.
5. Action Buttons (Donate / Volunteer / Info) – Enables users to directly donate, sign up as a volunteer, or learn more about the NGO.

VI. DATA ANALYSIS: Based on the survey

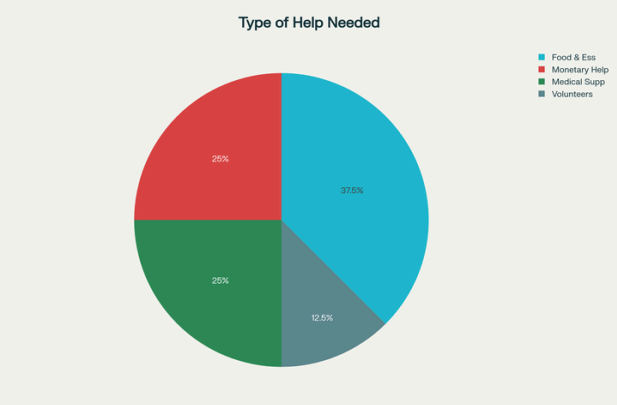


fig A

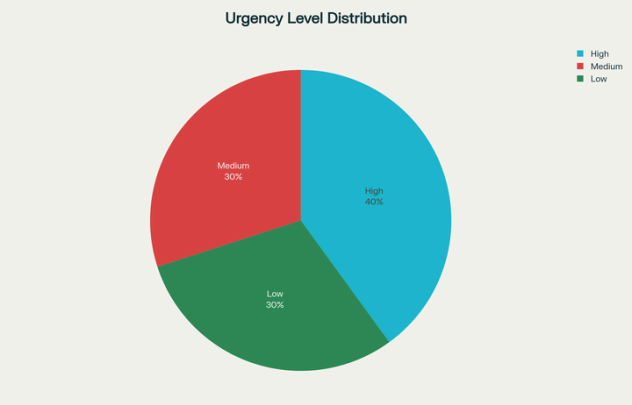


fig B

According to fig A:

The pie chart titled "Type of Help Needed" presents the following breakdown of requirements among sampled Pune NGOs:
Food & Essentials: 37.5%
Monetary Help: 25%
Medical Supplies: 25%
Volunteers: 12.5%

Each segment is color-coded and percentage-labeled, alongside a legend that facilitates easy interpretation. The proportions reflect a diverse set of organizational priorities and reflect evolving ground realities, such as economic hardship, food insecurity, public health crises, and volunteer workforce demand.

Analysis & Discussion

- Food & Essentials Dominance: The largest need is for food and everyday essentials, representing over a third of requests. This suggests a continued prevalence of poverty and food insecurity among the populations these NGOs serve. As such, resource mobilization for food relief must remain a high priority.
- Monetary & Medical Aid: At 25% each, requests for financial support and medical supplies are equally prominent. Medical needs may reflect both routine health care delivery and heightened pandemic recovery efforts. Financial assistance, on the other hand, remains critical for operational sustainability and expansion of direct service programs.
- Volunteer Support: While still significant, the demand for volunteers constitutes the smallest segment (12.5%). This points to efficient utilization or existing sufficiency of human resources for some NGOs, or possibly structural constraints on volunteer deployment.

According to fig B:

The chart divides requests into three distinct urgency categories:
High Urgency: 40%
Medium Urgency: 30%
Low Urgency: 30%

Each segment is clearly color-coded and percentage-labeled, supported by a legend to facilitate rapid interpretation. This distribution captures the relative prevalence of critical versus routine needs across 10 sampled NGOs operating in Pune.

Analysis & Discussion

- Prevalence of High Urgency: A notable 40% of requests fall into the "High" urgency category, indicating that a substantial portion of the NGO community is grappling with acute challenges. These could include health emergencies, food insecurity spikes, disaster relief, or other crisis situations requiring immediate intervention.
- Medium and Low Urgency: Requests tagged as "Medium" and "Low" each comprise 30% of the total. Medium urgency likely encompasses operational shortfalls, intermediate-level medical cases, or seasonal needs. Low urgency includes requests where planned support or regular programming suffices.
- Resource Allocation Implications: The data suggests that while there are ongoing and future support needs, a relatively large proportion of NGO requests must be addressed as a priority. This justifies a responsive allocation model in which high-urgency requests receive a disproportionately greater share of funds, volunteer hours, and in-kind donations.

VII. RESULT AND IMPACT

Accelerated Response: By improving visibility of needs and providing a simple well-organized dashboard for incoming requests, *Helping Hand* significantly shortens the gap between when help is asked for and when it is delivered. This allows vulnerable communities to receive support at the right time, which is specially critical during emergencies or prolonged hardships, helping reduce the suffering and in many cases save lives

Empowered Small Initiatives: The platform creates opportunities for the local NGOs and individual helpers who often go unnoticed on larger platforms. By giving these initiatives equal visibility, *Helping Hands* encourages stronger community participation and motivates more donors and volunteers to step forward and contribute

Efficient distribution: *Helping Hands* relies on data-driven methods to connect available resources with real needs. This approach reduces overlap, limits unnecessary waste and ensure the assistance reaches those who need it the most. As a result, aid efforts become more effective and financially efficient

Fostering trust: Trust is strengthened through transparency. By clearly displaying request updates, verified profiles and shared success stories, the platform builds confidence among donors, volunteers and recipients alike. This sense of trust encourages ongoing cooperation and consistent support within the helping community.

VII. CONCLUSION

Helping hands goes beyond being a simple digital platform; it serves as a meaningful bridge between compassionate individuals and organizations working towards social good. By combining transparency, thoughtful design and scalable technology, the platform makes it easier for the help to reach those who truly need it. As the system continues to evolve, it will become more and even more effective in supporting marginalized and often overlooked communities, ensuring that genuine requests for assistance are not ignored. Through the creation of an active and caring network of donors, volunteers and beneficiaries, *Helping Hands* highlights how technology, when used responsibly, can strengthen humanitarian efforts and create lasting social impact.

IX. FUTURE SCOPE

Helping Hands plans to keep evolving and growing its abilities for a wider impact. Upcoming improvements include:

Donor Login and Contribution Tracking: *Helping Hands* plans to introduce secure donor accounts that allows users to login, track contributions and receive regular updates on how support is being used. This feature strengthens accountability and helps donors feel more personally connected to the causes they support.

Advanced Request Categorization: The platform will add detailed categories based on aid type, urgency and beneficiary demographics. These refinements improve recommendation accuracy, reduce delays and ensure resources are directed to the right organizations at the right time.

Native Mobile Applications: Dedicated Android and iOS applications will make participation easier and more immediate. Mobile access enables faster responses to urgent needs and supports users in areas with limited desktops availability.

NGO and Helper Profile Verification: Strong verification processes will confirm authenticity of NGOs and helpers. This increases reliability, builds trust and creates a safer environment for meaningful humanitarian collaboration for all platform users.

These upgrades aim to change Helping Hands from a simple matching platform into a full ecosystem for coordinated humanitarian aid.

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